

The Twenty-Seven-Line Field of the Fourth Hénon Yukawa Surface

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Abstract

We determine the arithmetic of the 27-line scheme of a specific smooth cubic surface over $\mathbf{Q}$, namely the surface cut out by the four-variable Yukawa cubic associated with the fourth Hénon core. An exact Grassmannian calculation produces a degree-27 eliminant and a quotient with standard monomial counts $(1, 4, 10, 12, 0)$. Direct back-substitution, followed by a scheme-theoretic open-and-closed argument, identifies the complete Fano scheme with $\mathrm{Spec}(E)$ for a degree-27 number field E . Four complete squarefree factorizations modulo 7, 19, 29, 37 have degree patterns

$$(3, 3, 3, 3, 3, 6, 6), \quad (1, 4, 4, 6, 12), \quad (1, 2, 8, 8, 8), \quad (2, 5, 5, 5, 10).$$

Their factor-degree subset sums intersect only in $\{0, 27\}$, proving that the eliminant is irreducible. The last Frobenius type, combined with an exact enumeration of the 51840-element Weyl group, excludes its index-two Coxeter-even subgroup and gives $\mathrm{Gal}(K/\mathbf{Q}) \cong W(E_6)$ for the normal closure K of E . Consequently the geometric and arithmetic Picard ranks are 7 and 1. No line is defined over $\mathbf{Q}$; more generally, every finite extension $L/\mathbf{Q}$ over which a line is defined has degree divisible by 27. The result concerns lines and Picard ranks only: it makes no assertion about rational points, rationality, Brauer–Manin obstructions, motives, or Calabi–Yau realizations.

1 Introduction

A smooth cubic surface has 27 geometric lines, but the base field need not see any of them individually. Their Galois action can split into several orbits, and even a transitive action need not be the full incidence group $W(E_6)$. For a fixed surface, determining the line field therefore requires more than smoothness or the classical line count.

This paper carries out that determination for the exact cubic surface $Y/\mathbf{Q}$ displayed in equation (1). The equation arose as a four-variable Yukawa cubic in the preceding Hénon-core construction, but the present problem is independent of the Hodge-theoretic origin. Its object is the finite scheme

$$F_1(Y) \subset \mathrm{Gr}(2, 4),$$

and its central questions are arithmetic: is this scheme connected, what is the normal field of its geometric points, and what does the resulting action fix in the Picard lattice?

Three points make the calculation delicate. First, a zero-dimensional calculation on one Grassmann chart does not automatically recover the global Fano scheme. We close that gap by first proving that $F_1(Y)$ is finite étale of rank 27; an open chart is then open-and-closed, and a rank-27 closed subscheme must be the whole line scheme. Second, a transitive subgroup containing an element of order five may still be the index-two simple subgroup $U \subset W(E_6)$. The required parity is the determinant in the E_6 reflection representation, not ordinary permutation sign: every element of $W(E_6)$ acts evenly on the 27 lines. Third, the degree-27 residue field of

one line is not the common normal line field. We denote the former by E and its normal closure by K throughout.

The main result is as follows.

Theorem 1.1 (Line field, Galois closure, and Picard ranks). *For the cubic surface $Y/\mathbf{Q}$ in equation (1), there is an irreducible polynomial $g \in \mathbf{Z}[d]$ of degree 27 such that*

$$F_1(Y) \cong \operatorname{Spec}(E), \quad E = \mathbf{Q}[d]/(g).$$

Thus $F_1(Y)$ is connected and finite étale of rank 27. If K is the splitting field of g , then

$$\operatorname{Gal}(K/\mathbf{Q}) \cong W(E_6), \quad [K : \mathbf{Q}] = 51840.$$

Moreover,

$$\rho(Y_{\overline{\mathbf{Q}}}) = 7, \quad \rho(Y/\mathbf{Q}) = 1.$$

If a finite extension $L/\mathbf{Q}$ defines a line on Y , then

$$27 \mid [L : \mathbf{Q}].$$

In particular, Y has no $\mathbf{Q}$ -rational line.

The proof combines written geometry and group theory with finite exact calculations. Its four principal contributions are:

1. We reconstruct the complete line scheme from one exact Grassmann chart. The degree-order quotient has dimension 27 and Hilbert counts $(1, 4, 10, 12, 0)$; a four-polynomial lexicographic shape gives a degree-27 eliminant and three linear back-substitutions. All four original line equations reduce to zero.
2. We prove irreducibility by complete modular data rather than a black-box rational factorization. The factor-degree subset-sum intersections shrink through

$$\{0, 3, 6, \dots, 27\}, \quad \{0, 6, 9, 12, 15, 18, 21, 27\}, \quad \{0, 9, 18, 27\}, \quad \{0, 27\}.$$

3. We prove maximal Galois action. The prime 37 gives cycle type $(2, 5, 5, 5, 10)$. Exact enumeration finds 5184 elements of this type in $W(E_6)$, all outside the Coxeter-even subgroup U , and none inside it. Together with the transitive order-five criterion of Elsenhans–Jahnel, this forces the full Weyl group [2].
4. We reconstruct the Picard reflection model and find a one-dimensional fixed space. Hochschild–Serre is used only after tensoring with $\mathbf{Q}$, so the conclusion is equality of ranks rather than an unqualified integral equality of Picard groups.

The arithmetic conclusion about lines is intentionally narrow. A surface without a rational line may still have rational points, and nothing here decides rationality, stable rationality, the Hasse principle, or a Brauer–Manin obstruction. The Yukawa origin supplies the distinguished equation but does not turn the line-field computation into a motivic or Calabi–Yau realization theorem.

The rest of the paper is organized as follows. Section 2 fixes the surface and establishes the global finite étale line scheme. Section 3 gives the chart presentation and the open-and-closed bridge. Section 4 proves that the degree-27 algebra is a field. Section 5 determines its normal closure, and section 6 derives the Picard and line-definition consequences. Exact replay and scope are recorded in sections 7 and 8; the appendices contain source locators and the compact exact data.

2 The fixed surface and its Fano scheme

2.1 The cubic

In homogeneous coordinates u_0, u_1, u_2, u_3 , let

$$\begin{aligned}
F = & 75081586157u_0^3 - 28576620789u_0^2u_1 - 122000922135u_0^2u_2 - 5364921951u_0^2u_3 \\
& + 164150208636u_0u_1^2 - 415458334296u_0u_1u_2 + 151070718312u_0u_1u_3 \\
& + 1158143874300u_0u_2^2 + 114691988016u_0u_2u_3 + 113572676646u_0u_3^2 \\
& + 6898957820u_1^3 + 1132596902196u_1^2u_2 - 30413540316u_1^2u_3 \\
& - 2054867641020u_1u_2^2 + 151980984216u_1u_2u_3 + 36794420832u_1u_3^2 \\
& + 2646295985484u_2^3 + 560186573940u_2^2u_3 + 706181383584u_2u_3^2 + 1884468968u_3^3, \quad (1)
\end{aligned}$$

and set $Y = V(F) \subset \mathbf{P}_{\mathbf{Q}}^3$. The coefficient content is one and the first coefficient is positive, which fixes the common-scalar convention. The current exact replay reconstructs all twenty sparse coefficient rows and verifies projective smoothness over $\mathbf{Q}$ by unit ideals on the four standard affine charts. It also verifies good reduction at 7, 19, 29, 37.

The equation is the only Hénon/Yukawa input used below. A rational coordinate transformation in $\mathrm{GL}_4(\mathbf{Q})$, or multiplication of F by an element of $\mathbf{Q}^\times$, changes its presentation but not any line-scheme or Picard conclusion; see corollary 6.3.

2.2 The line section

Let $\mathcal{S}$ be the tautological rank-two subbundle on $\mathrm{Gr}(2, 4)$. Restriction of F to a two-plane defines a section

$$\sigma_F \in H^0\left(\mathrm{Gr}(2, 4), \mathrm{Sym}^3(\mathcal{S}^\vee)\right), \quad [S] \mapsto F|_S. \quad (2)$$

Its zero scheme is the Fano scheme $F_1(Y)$. This is exactly the construction in Kass–Wickelgren, Definition 41 [3].

Two distinct external inputs control its geometric size and reducedness. The rank of the arithmetic count in Kass–Wickelgren, Theorem 2, is $15 + 12 = 27$, recovering the total geometric degree 27 for a smooth cubic surface. Corollary 53, on the other hand, says that the zero of σ_F corresponding to every line disjoint from the singular locus is simple. Since Y is smooth, every line satisfies that condition. Corollary 54 separately records separability of the residue fields. We do not attribute the total count 27 to Corollary 53.

Proposition 2.1. *The scheme $F_1(Y)$ is finite étale over $\mathbf{Q}$ of rank 27.*

Proof. The zero scheme of σ_F is closed in the projective Grassmannian. Over $\overline{\mathbf{Q}}$, the total-degree statement gives 27 zeros and Kass–Wickelgren Corollary 53 makes every zero simple. Thus $F_1(Y)_{\overline{\mathbf{Q}}}$ is a disjoint union of 27 reduced points. Consequently $F_1(Y)$ is zero-dimensional and proper, hence finite. Geometric reducedness over the characteristic-zero field $\mathbf{Q}$ makes the finite scheme étale, with rank 27. $\square$

Remark 2.2. Proposition 2.1 is proved before any affine chart is used. This order is needed in section 3: it makes every chart open in $F_1(Y)$ also closed, preventing a hidden assumption that one chart already contains all lines.

3 A degree-27 chart presentation

3.1 The U_{01} equations

On the standard Plücker chart U_{01} , write a line as the row span of

$$M(a, b, c, d) = \begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{pmatrix}. \quad (3)$$

With parameters s, t , its points are

$$(s, t, as + ct, bs + dt).$$

Restricting equation (1) gives

$$F(s, t, as + ct, bs + dt) = f_0 s^3 + f_1 s^2 t + f_2 s t^2 + f_3 t^3. \quad (4)$$

The four sparse polynomials $f_i \in \mathbf{Q}[a, b, c, d]$ are printed in appendix B. Their common zero scheme is $F_1(Y) \cap U_{01}$.

The degree-order calculation has a 21-element Gröbner basis. Its standard monomials have total-degree counts

$$(H_0, H_1, H_2, H_3, H_{\geq 4}) = (1, 4, 10, 12, 0), \quad (5)$$

so the quotient dimension is 27. Exact FGLM conversion gives a four-element lexicographic shape

$$\begin{aligned} g(d) &= 0, \\ \lambda_c c + h_c(d) &= 0, \\ \lambda_b b + h_b(d) &= 0, \\ \lambda_a a + h_a(d) &= 0, \end{aligned} \quad \lambda_a \lambda_b \lambda_c \neq 0, \quad (6)$$

where $g \in \mathbf{Z}[d]$ is primitive of degree 27, and each $h_\bullet$ has degree at most 26. The complete coefficient arrays are carried by the exact electronic certificate; the compact invariants and the coefficient endpoints of g are recorded in appendix B.

The shape computation alone is not used as an ideal-membership proof. Instead, the independent replay substitutes

$$a = -\frac{h_a(d)}{\lambda_a}, \quad b = -\frac{h_b(d)}{\lambda_b}, \quad c = -\frac{h_c(d)}{\lambda_c} \quad (7)$$

into each f_i , clears the three constant denominators, and reduces the resulting numerator modulo g . All four remainders are exactly zero.

3.2 From the chart to the global scheme

Let

$$A_g = \mathbf{Q}[d]/(g).$$

The four direct remainder identities induce a surjection

$$\mathbf{Q}[a, b, c, d]/(f_0, f_1, f_2, f_3) \twoheadrightarrow A_g, \quad (8)$$

because the image contains the class of d . Dually, this gives a closed immersion

$$Z := \text{Spec}(A_g) \hookrightarrow F_1(Y) \cap U_{01}. \quad (9)$$

Lemma 3.1 (Open-and-closed bridge). *The immersion in equation (9) is a closed immersion into the global scheme $F_1(Y)$.*

Proof. By proposition 2.1, $F_1(Y)$ is a finite étale scheme over a field. Its underlying space is finite and discrete, and every open subscheme is a union of connected components. Thus $F_1(Y) \cap U_{01}$ is open-and-closed in $F_1(Y)$. Composing equation (9) with this closed inclusion proves the claim. $\square$

Proposition 3.2 (Complete chart presentation). *There is an isomorphism*

$$F_1(Y) \cong \operatorname{Spec}(\mathbf{Q}[d]/(g)).$$

In particular all 27 geometric lines lie in U_{01} .

Proof. The algebra A_g has $\mathbf{Q}$ -dimension 27, because $\deg g = 27$. Hence Z and $F_1(Y)$ are finite schemes of the same rank. The global closed immersion from lemma 3.1 corresponds to a surjective map between their 27-dimensional coordinate algebras. It is therefore an isomorphism. $\square$

3.3 Independent complement guard

The exact replay also generates the other five standard charts and adds the equation $p_{01} = 0$, which cuts out the complement of U_{01} . Each resulting ideal has reduced Gröbner basis $\{1\}$:

chart	row-coordinate convention	equation for $p_{01} = 0$
U_{02}	$(s, as + ct, t, bs + dt)$	$c = 0$
U_{03}	$(s, as + ct, bs + dt, t)$	$c = 0$
U_{12}	$(as + ct, s, t, bs + dt)$	$c = 0$
U_{13}	$(as + ct, s, bs + dt, t)$	$c = 0$
U_{23}	$(as + ct, bs + dt, s, t)$	$ad - bc = 0$

This calculation independently audits the chart convention. It is not used circularly to justify lemma 3.1; global equality already follows from finite étale rank comparison.

4 Four-prime irreducibility

Proposition 3.2 identifies the full line scheme with a degree-27 finite $\mathbf{Q}$ -algebra. To prove connectedness, it remains to show that g is irreducible.

Lemma 4.1 (Subset-sum obstruction). *Let $g \in \mathbf{Z}[d]$ be primitive of degree n . Suppose that, for a prime p , its leading coefficient survives modulo p and*

$$\bar{g} = u_p \prod_{j=1}^{r_p} q_{p,j}$$

is a squarefree factorization into distinct monic irreducibles of degrees $e_{p,1}, \dots, e_{p,r_p}$. If g has a rational factor of degree m , then

$$m \in S_p := \left\{ \sum_{j \in J} e_{p,j} : J \subseteq \{1, \dots, r_p\} \right\}.$$

Proof. Gauss's lemma supplies a primitive integral factorization $g = vw$ with $\deg v = m$. Since the leading coefficient of $\bar{g}$ is nonzero, reduction preserves the degrees of both factors. Squarefreeness makes $\bar{v}$ a product of a subset of the distinct $q_{p,j}$, so its degree is a subset sum of their degrees. $\square$

4.1 Complete modular witnesses

The exact factorizations are squarefree, preserve degree, and multiply back to the reduction of g . Their compact data are:

p	leading coefficient	factorization unit	factor degrees
7	2	2	(3, 3, 3, 3, 6, 6)
19	8	8	(1, 4, 4, 6, 12)
29	26	26	(1, 2, 8, 8, 8)
37	29	29	(2, 5, 5, 5, 10)

All multiplicities are one, and in each case $\gcd(\bar{g}, \bar{g}') = 1$. The monic factors themselves are listed in appendix C; recording only their degrees would not certify factor multiplication or squarefreeness.

The corresponding subset-sum sets are

$$\begin{aligned}
S_7 &= \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27\}, \\
S_{19} &= \{0, 1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 26, 27\}, \\
S_{29} &= \{0, 1, 2, 3, 8, 9, 10, 11, 16, 17, 18, 19, 24, 25, 26, 27\}, \\
S_{37} &= \{0, 2, 5, 7, 10, 12, 15, 17, 20, 22, 25, 27\}.
\end{aligned} \tag{10}$$

Their successive intersections are

$$\begin{aligned}
S_7 &= \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27\}, \\
S_7 \cap S_{19} &= \{0, 6, 9, 12, 15, 18, 21, 27\}, \\
S_7 \cap S_{19} \cap S_{29} &= \{0, 9, 18, 27\}, \\
S_7 \cap S_{19} \cap S_{29} \cap S_{37} &= \{0, 27\}.
\end{aligned} \tag{11}$$

Proposition 4.2. *The polynomial g is irreducible over $\mathbf{Q}$. Consequently*

$$E := \mathbf{Q}[d]/(g)$$

is a degree-27 number field and

$$F_1(Y) \cong \text{Spec}(E)$$

is connected and finite étale.

Proof. If g had a proper rational factor of degree m , then $1 \leq m \leq 26$. Applying lemma 4.1 at all four primes would put m in the last intersection of equation (11), which is $\{0, 27\}$. This is impossible. The remaining assertions follow from proposition 3.2. $\square$

Remark 4.3. The proof uses the complete modular factorizations as exact identities. A rational CAS declaration that g is irreducible is neither needed nor treated as an independent proof.

5 The normal line field and the full Weyl group

5.1 The two fields

Let K be the splitting field of the irreducible eliminant g . The 27 embeddings $E \hookrightarrow \overline{\mathbf{Q}}$ send the class of d to the 27 roots of g , hence to the 27 geometric points of $F_1(Y)$. The rational back-substitutions in equation (7) recover a, b, c from each root. Thus K contains the coordinates of every line.

Conversely, a normal extension over which all 27 lines are defined contains all of their d -coordinates and therefore every root of g . It contains K . Hence K is the least normal common field of definition of the lines and is the normal closure of E .

This distinction will matter numerically:

$$[E : \mathbf{Q}] = 27, \quad [K : \mathbf{Q}] = 51840.$$

In particular $E \neq K$.

5.2 Transitivity and the order-five gate

Set

$$G = \text{Gal}(K/\mathbf{Q}).$$

The action of G on the roots is faithful by definition of a splitting field and transitive by proposition 4.2. The intersections among the reconstructed lines are defined geometrically and are preserved by Galois. Elsenhans–Jahnel Fact 3 therefore embeds G into the Weyl group $W(E_6)$ acting on the Schläfli configuration [2].

At $p = 37$, the leading coefficient survives and the complete factorization is squarefree. The prime is therefore unramified in the root permutation, and the factor degrees give a Frobenius element

$$\varphi \in G \subseteq W(E_6) \quad \text{with cycle type} \quad (2, 5, 5, 5, 10). \quad (12)$$

Its order is 10, and φ^2 has order 5. Elsenhans–Jahnel Lemma 8 applies to the transitive group G : it leaves exactly the alternatives

$$G = U \quad \text{or} \quad G = W(E_6), \quad (13)$$

where U is the simple index-two subgroup of order 25920. The order-five argument alone does not choose between them.

5.3 The parity firewall

The subgroup U is the kernel of determinant on the six-dimensional E_6 reflection representation. Equivalently, it is the subgroup of even Coxeter word parity. This is not ordinary permutation parity on the 27 lines. Elsenhans–Jahnel Remark 5 states that the entire image of $W(E_6)$ on those lines lies in A_{27} . In particular,

$$\text{sgn}_{S_{27}}(w) = +1 \quad \text{for every } w \in W(E_6). \quad (14)$$

The target type in equation (12) also has ordinary sign +1; ordinary sign cannot exclude U .

The exact Picard-lattice enumeration constructs $W(E_6)$ from its six simple reflections and obtains:

quantity	exact value
elements of $W(E_6)$	51840
elements of U	25920
orbit size on line classes	27
elements of type $(2, 5, 5, 5, 10)$	5184
target-type elements in U	0
target-type elements outside U	5184
elements acting oddly in S_{27}	0

Thus every element of the Frobenius type lies outside U . This instance computation agrees with Elsenhans–Jahnel Algorithm 10 and Remarks 11–13: the pattern $(2, 5, 5, 5, 10)$ is one of their selected outside- U classes and simultaneously supplies an order-five element.

Theorem 5.1 (Maximal line-field Galois group). *The common normal line field K satisfies*

$$\mathrm{Gal}(K/\mathbf{Q}) \cong W(E_6), \quad [K : \mathbf{Q}] = 51840.$$

The degree-27 field E is not Galois over $\mathbf{Q}$.

Proof. The Frobenius element φ lies outside U , so the first alternative in equation (13) is impossible. Hence $G = W(E_6)$, whose exact order is 51840. If $E/\mathbf{Q}$ were Galois, its normal closure would equal E and have degree 27, contradicting the degree of K . $\square$

6 Picard ranks and fields defining a line

6.1 The geometric Picard lattice

Over $\overline{\mathbf{Q}}$, a smooth cubic surface is the blow-up of $\mathbf{P}^2$ at six points. Write

$$\mathrm{Pic}(Y_{\overline{\mathbf{Q}}}) = \mathbf{Z}H \oplus \mathbf{Z}E_1 \oplus \cdots \oplus \mathbf{Z}E_6,$$

with intersection form

$$\mathrm{diag}(1, -1, -1, -1, -1, -1, -1). \quad (15)$$

Thus $\rho(Y_{\overline{\mathbf{Q}}}) = 7$. The 27 line classes are

$$E_i, \quad H - E_i - E_j \ (i < j), \quad 2H - \sum_{j \neq i} E_j. \quad (16)$$

The six roots

$$E_1 - E_2, \ E_2 - E_3, \ E_3 - E_4, \ E_4 - E_5, \ E_5 - E_6, \ H - E_1 - E_2 - E_3 \quad (17)$$

all have self-intersection -2 . For such a root α , the reflection is

$$s_\alpha(x) = x + (x \cdot \alpha)\alpha. \quad (18)$$

The exact replay verifies that these six maps preserve equation (15), the 27 classes in equation (16), and their full intersection matrix. They generate a group of order 51840, act transitively on the line classes, and have determinant -1 on the E_6 root space.

The anticanonical class

$$-K_Y = 3H - E_1 - \cdots - E_6$$

is fixed by every reflection. The six roots span its orthogonal complement. Therefore a rational Picard vector fixed by all reflections is a multiple of $-K_Y$, and

$$\mathrm{rank} \left(\mathrm{Pic}(Y_{\overline{\mathbf{Q}}}) \otimes \mathbf{Q} \right)^{W(E_6)} = 1. \quad (19)$$

The independent matrix calculation gives the same fixed rank by stacking the six matrices $s_\alpha - I$.

6.2 Hochschild–Serre at the rank level

By theorem 5.1, the Galois image on the geometric Picard lattice is the full group $W(E_6)$. Hence equation (19) gives

$$\mathrm{rank} \mathrm{Pic}(Y_{\overline{\mathbf{Q}}})^{G_{\mathbf{Q}}} = 1.$$

The low-degree Hochschild–Serre sequence, together with Hilbert 90, contains

$$0 \longrightarrow \mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(Y_{\overline{\mathbf{Q}}})^{G_{\mathbf{Q}}} \longrightarrow \mathrm{Br}(\mathbf{Q}), \quad (20)$$

as recorded, for example, in Viray, §2.1, equation (24) and the following exact sequence [6]. Since $\text{Br}(\mathbf{Q})$ is torsion, the cokernel of the Picard injection is torsion. Tensoring equation (20) with $\mathbf{Q}$ therefore gives equality of ranks. Moreover, $\text{Pic}^0(Y_{\overline{\mathbf{Q}}}) = 0$ because a smooth cubic surface is geometrically rational. Thus its Picard group equals its Néron–Severi group, and the common rank is the arithmetic Picard number:

$$\rho(Y/\mathbf{Q}) = \text{rank Pic}(Y) = \text{rank Pic}(Y_{\overline{\mathbf{Q}}})^{G_{\mathbf{Q}}} = 1. \quad (21)$$

We do not assert an integral equality $\text{Pic}(Y) = \text{Pic}(Y_{\overline{\mathbf{Q}}})^{G_{\mathbf{Q}}}$.

6.3 Line-definition fields

Corollary 6.1. *Let $L/\mathbf{Q}$ be finite. If Y has a line defined over L , then*

$$27 \mid [L : \mathbf{Q}].$$

In particular Y has no $\mathbf{Q}$ -rational line.

Proof. An L -defined line is an L -point of $F_1(Y)$. By proposition 4.2, this is a $\mathbf{Q}$ -algebra map $E \rightarrow L$. Because E is a field, the map is injective. Its image is a conjugate $E' \subset L$ with $[E' : \mathbf{Q}] = 27$, and the tower law gives

$$[L : \mathbf{Q}] = [L : E'] [E' : \mathbf{Q}] = 27 [L : E'].$$

□

Remark 6.2 (No rationality inference). Corollary 6.1 concerns rational lines, not rational points. It proves neither $Y(\mathbf{Q}) = \emptyset$ nor any statement about rationality, stable rationality, the Hasse principle, or a Brauer–Manin obstruction.

Corollary 6.3 (Projective invariance). *A rational projective coordinate change and multiplication of F by an element of $\mathbf{Q}^\times$ preserve the isomorphism class of $F_1(Y)$, the fields E and K up to $\mathbf{Q}$ -isomorphism, the Galois permutation representation on the 27 lines, and both Picard ranks.*

Proof. A coordinate change transports the line section equation (2) along the induced Grassmannian automorphism, while a common nonzero scalar leaves its zero scheme unchanged. The remaining objects and ranks are functorial consequences of that finite scheme and its incidence configuration. □

7 Exact replay and evidence boundary

The proof separates three kinds of evidence. General theorems control the geometry of line sections and the subgroup alternatives. Exact computation supplies the large instance-specific identities. Written arguments connect those inputs to the global scheme, Galois, and Picard conclusions.

7.1 Claim-to-evidence ledger

Claim	Exact machine input	Written or external bridge
$F_1(Y)$ finite étale, rank 27	exact smoothness replay	Kass–Wickelgren Theorem 2 gives total rank; Corollary 53 gives simple zeros
global chart equality	quotient dimension 27, lex shape, four direct zero remainders	finite étale makes the chart open-and-closed; equal rank gives equality
g irreducible	four complete squarefree modular factorizations and subset sums	Gauss’s lemma and lemma 4.1
$\text{Gal}(K/\mathbf{Q}) = W(E_6)$	prime-37 factorization and exact Weyl/Coxeter enumeration	Frobenius cycle type, Elsenhans–Jahnel Lemma 8, exclusion of U
$\rho(Y/\mathbf{Q}) = 1$	fixed rank 1 for the explicit $W(E_6)$ -action on the geometric Picard lattice	Hochschild–Serre torsion cokernel, followed by tensoring with $\mathbf{Q}$
no $\mathbf{Q}$ -line	connected degree-27 Fano scheme	the universal property of $\text{Spec}(E)$ and the tower law

The Picard row is a release-critical boundary. The machine calculation does *not* directly certify the arithmetic Picard rank. It certifies the rank-one fixed subspace of the geometric Picard representation. The passage to $\rho(Y/\mathbf{Q}) = 1$ is the written Hochschild–Serre argument in equations (20) and (21).

7.2 Independent finite gates

The producer reconstructs the frozen twenty-term cubic, generates the six Grassmann chart systems, computes the degree-order and lexicographic bases, factors the eliminant at the four selected primes, and constructs the Weyl group from Picard reflections. It emits a strict canonical JSON payload.

The checker imports no producer module. It independently:

1. validates the layered frozen C55 source contract before importing the cubic;
2. rebuilds the surface and all chart equations;
3. recomputes the standard monomial box and all four direct remainders;
4. verifies the five complementary unit ideals;
5. multiplies every stored modular factor, checks derivative gcds, and recomputes every subset-sum set;
6. rebuilds the Picard lattice, the 27 line classes, all six reflections, the 51840-element group, Coxeter kernel, target class counts, and fixed rank;
7. derives each terminal theorem gate from those data rather than trusting a producer conclusion string.

The schema rejects unknown or missing keys, duplicate JSON keys, floats, booleans in integer fields, noncanonical integers, oversized input, and optimized execution that would suppress assertions. The hostile suite mutates every classified scalar leaf while rebinding the exposed envelope digests; semantic leaves must still be rejected for mathematical reasons. It also exercises source drift, symlink and path attacks, stale outputs, promotion-order failures, rollback after every move, and byte/mtime nonmutation in read-only check mode.

7.3 Official code/results prefreeze tuple

The refreshed prefreeze payload encodes the machine fixed rank and the requirement for the written Hochschild–Serre bridge as separate leaves. A fresh default replay passes all ten semantic gates. The exact code/results prefreeze tuple used by this source draft is:

field	prefreeze value
semantic gates	10/10
semantic rebound sweep	2684/2684 = 2662 payload + 20 schema + 2 envelope
test methods	15/15
payload SHA-256	5b17c9ed7bea60680556af70297199b653d51188bb30ce59f7c2c6bfb9c94f661
certificate SHA-256	26739ce5aedb4a3467645f9c1b2036d4d3e9ce4d0dbce23d67ea7b67e5fbc4
canonical schema SHA-256	ef26d7204a38e28aaf00eed8188b31d34d590c9c8a19924f1d0798e40b052d5f
schema-file SHA-256	adab34998a944c8a4af8db774e511f0453839ea6a6e14e9437ffc259be3da504
independent-check SHA-256	4ccfb09139a4bfa812ea9c57ff8b65a6a8e603dbdb00e245355a4563386489a9
scoped-manifest SHA-256 and entry count	20d29af97128e766bb5e59bf6f82f8401c6ed62f279371b031febcefd5d99b4a; 12
producer/checker/test direct release fields	bound by the scoped manifest; otherwise null
implementation/provenance commits	null
full-project successor	root FULL_PROJECT_HASHES.sha256; self-excluding 46 entries; verified separately; digest external-only

The self-excluding scoped manifest remains the default code/results identity and default runner scope. The code/results state is `PREFREEZE_CODE_RESULTS_PASS`, while the no-commit project is a `RELEASE_CANDIDATE`. The separately verified root successor covers the 46-entry release-wide tree without replacing that scoped identity; its digest is external-only to avoid self-reference. Temporary architecture, reconnaissance, and hostile-review digests are chronology only and are never theorem inputs.

7.4 Computational boundary

Computer algebra certifies finite identities: coefficients, ideal membership, quotient dimensions, modular products, group enumeration, and matrix ranks. It does not supply the classical line theorem, the open-and-closed scheme argument, the subgroup theorem, the Hochschild–Serre rank bridge, or the universal property used for fields of definition. Conversely, the cited theorems do not verify the eliminant, factors, or Weyl enumeration for this surface. Both layers are needed.

A controlled bootstrap live build completed after the independent isolated source audit. It produced an (18)-page A4 PDF with zero final-pass TeX/BibTeX warnings, box warnings, unresolved references or citations, duplicate destinations, and rerun requests; all (26) fonts were embedded, subsetted Type 1 fonts. Its diagnostic predecessor identifiers were:

bootstrap PDF SHA-256	241d631aad89aea4a8885fb36187b8509ce2316a51d19ef6b51a82824c478f0c
bootstrap log SHA-256	9a6595a90af1bf5f8091ded907aab00721e4542cbfda9944fb20550e21e8df30
bootstrap text SHA-256	037a5dd42a69b2f0219af912f96ca4629628c09306bbd0c427d189b6f4729c91.

These three bootstrap digests are chronology-only and are not release authority. The controlled final rebuild supersedes them; its paper-source, PDF, log, extracted-text, and compilation-report digests are recorded externally in the Route record and compilation report, rather than embedded here, to avoid a self-hash cycle.

8 Context, novelty boundary, and limitations

The 27-sheeted incidence problem. The moduli space of smooth complex cubic surfaces has a natural incidence cover obtained by marking a line. Das studies this cover and its topology [1]. Our object is one arithmetic fibre of that general construction, presented as an explicit degree-27 field rather than as a family-level topological cover.

Arithmetic line counts. The possible numbers of rational lines and their dependence on the base field have a long classical history. McKean gives a modern field-theoretic treatment using subgroups acting on the Schläfli graph [4]. Here the absence of a rational line is not inferred from a classification table: it follows directly because the entire line scheme is the spectrum of one degree-27 field.

Galois computations and symmetry. Elsenhans–Jahnel develop modular tests for Galois groups of cubic surfaces, including the transitive order-five criterion used in section 5 [2]. Recent work of Pichon–Pharabod–Telen studies monodromy groups in symmetric linear systems and finds that symmetry often forces proper subgroups of $W(E_6)$ [5]. That nearby result makes a maximality calculation essential for the present distinguished surface; symmetry or provenance cannot substitute for the outside- U witness.

Bounded novelty statement. On 2026-08-15 we screened exact-title and arXiv queries combining the terms Yukawa cubic surface, 27 lines, Galois, and $W(E_6)$, together with recent 2024–2026 cubic-surface neighbors. The bounded search did not locate a prior computation of the four-prime degree-27 line field and full Weyl closure for the exact twenty-term surface in equation (1). This statement is limited by the listed query families, indexed sources, and search date. We do not claim exhaustive priority or use priority language.

8.1 What the theorem does not decide

The result has four deliberate boundaries.

1. A rational line is a special rational point of the Grassmannian Fano scheme. Its absence says neither that $Y(\mathbf{Q})$ is empty nor that Y is nonrational.
2. The torsion target in equation (20) is used only to compare Picard ranks. We compute no Brauer class and claim no Hasse-principle or Brauer–Manin obstruction.
3. The full line-field group is an instance result for equation (1). It is not a generic theorem for Yukawa, Hénon, symmetric, or arbitrary cubic surfaces.
4. The Hodge-theoretic origin of F does not turn an agreement of line schemes, Picard ranks, or Galois groups into a motive, a polarized variation of Hodge structure, or an honest Calabi–Yau realization.

The theorem is also projective: replacing the tangent-coordinate basis by an element of $\mathrm{GL}_4(\mathbf{Q})$ changes the displayed chart and eliminant but not the finite line scheme, its normal closure, or its Picard representation.

9 Conclusion and declarations

9.1 Conclusion

The line scheme of the fourth Hénon Yukawa cubic is one connected finite étale point of degree 27. Its normal closure has the largest Galois group compatible with cubic-surface incidence, namely $W(E_6)$ of order 51840. The same action leaves only the anticanonical line in the rational geometric Picard space; the Hochschild–Serre rank bridge then gives Picard ranks 7/1. Every finite extension $L/\mathbf{Q}$ over which a line is defined has degree divisible by 27.

The proof is designed so that no single computational shortcut carries a global conclusion. Direct remainders certify chart membership, finite étale geometry supplies the clopen bridge, complete modular factors prove irreducibility, Coxeter determinant excludes the index-two

subgroup, and a written descent argument converts geometric fixed rank to arithmetic Picard rank. The documentation build and provenance audit close only a no-commit release candidate; they add no new mathematical implication.

9.2 Declarations

Data and code availability. No external dataset, numerical fit, or trained parameter is used. The exact producer, independent checker, mutation suite, certificate, and compact scoped manifest are maintained in the HCS-C56 project. That 12-entry manifest remains the default code/results identity. A separate self-excluding 46-entry root successor records the release-wide tree and is verified externally; its digest is not embedded here. The current prefreeze identifiers are recorded in section 7; no temporary digest is a substitute.

Primary-source audit. Bibliographic metadata and theorem locators were checked against the DOI, publisher, author, and arXiv records listed in appendix A. The recent-neighbor screen is bounded rather than systematic or exhaustive.

Computational boundary. Machine calculations establish finite exact algebra and adversarial certificate behavior. The clopen scheme bridge, modular subset-sum lemma, subgroup implication, Hochschild–Serre rank step, and field-degree corollary are written mathematical arguments.

Author contributions. The work was carried out collaboratively: conceptualization, formal analysis, exact verification design, source control, writing, and curation.

Competing interests. The authors declare no competing interests.

Funding. No external funding is declared for this project.

Ethics. The work uses no human participants, animal subjects, or personal data.

A Primary-source locator ledger

The proof uses external sources only for general statements. Every instance-specific polynomial, factorization, group count, and matrix rank is supplied by the exact C56 replay.

source	exact locator	use and boundary
Kass–Wickelgren [3]	Theorem 2; §5, Definition 41; Corollary 53, p. 701; Corollary 54, p. 702	Definition 41 identifies the zero scheme of the $\mathrm{Sym}^3(\mathcal{S}^\vee)$ section with the line scheme. Theorem 2 has rank 27. Corollary 53 gives simple zeros on a smooth cubic, and Corollary 54 gives separable residue fields. Corollary 53 alone is not cited for the number 27.
Elsenhans–Jahnel [2]	Fact 3; Remarks 4–5; Lemma 8; Algorithm 10; Remarks 11–13; author-PDF pp. 3–7	Fact 3 places the common line-field group in $W(E_6)$. Lemma 8 reduces a transitive subgroup containing order five to U or $W(E_6)$. Remarks 11–13 certify the role of type $(2, 5, 5, 5, 10)$. Remark 5 says the 27-line action lies in A_{27} ; ordinary S_{27} sign does not define U .

source	exact locator	use and boundary
Viray [6]	§2.1, p. 10, equation (24) and the immediately following low-degree exact sequence	Supplies the Picard-to-Brauer segment of Hochschild–Serre. Since the field Brauer group is torsion, it yields rank equality after tensoring with $\mathbf{Q}$, not integral surjectivity and not a Brauer–Manin conclusion.
Das [1]	Introduction and the incidence cover $\widetilde{\mathcal{M}} \rightarrow \mathcal{M}$	Context for the 27-sheeted moduli cover that marks one line. It does not determine the arithmetic fibre studied here.
Pichon-Pharabod–Telen [5]	Abstract and §1 around Theorem 2 and Table 1	Recent context showing that symmetric cubic-surface families often have proper monodromy subgroups. It is not an exact line-field calculation for equation (1).
McKean [4]	Theorems 1.1–1.2, §1.2, and Appendix A	Context for rational-line counts and subgroup actions on the Schläfli graph. The no-line conclusion here instead follows from $F_1(Y) = \text{Spec}(E)$.

The Elsenhans–Jahnel chapter belongs to Progress in Mathematics volume 269; the publisher record carries copyright year 2009 and electronic publication in 2010. The bibliography uses the volume year 2009. The remaining published metadata were checked against the DOI records, and the preprint metadata against the official arXiv records, on 2026-08-15.

The recent-neighbor search used the query families listed in section 8. No temporary download path or digest is a bibliographic or theorem authority.

B Chart equations and eliminant data

B.1 The four restricted equations

For the chart convention in equation (3), direct expansion gives

$$\begin{aligned}
f_0 &= 2646295985484a^3 + 560186573940a^2b + 1158143874300a^2 \\
&\quad + 706181383584ab^2 + 114691988016ab - 122000922135a \\
&\quad + 1884468968b^3 + 113572676646b^2 - 5364921951b + 75081586157, \\
f_1 &= 7938887956452a^2c + 560186573940a^2d - 2054867641020a^2 \\
&\quad + 1120373147880abc + 1412362767168abd + 151980984216ab \\
&\quad + 2316287748600ac + 114691988016ad - 415458334296a \\
&\quad + 706181383584b^2c + 5653406904b^2d + 36794420832b^2 \\
&\quad + 114691988016bc + 227145353292bd + 151070718312b \\
&\quad - 122000922135c - 5364921951d - 28576620789, \\
f_2 &= 7938887956452ac^2 + 1120373147880acd - 4109735282040ac \\
&\quad + 706181383584ad^2 + 151980984216ad + 1132596902196a \\
&\quad + 560186573940bc^2 + 1412362767168bcd + 151980984216bc \\
&\quad + 5653406904bd^2 + 73588841664bd - 30413540316b \\
&\quad + 1158143874300c^2 + 114691988016cd - 415458334296c \\
&\quad + 113572676646d^2 + 151070718312d + 164150208636, \\
f_3 &= 2646295985484c^3 + 560186573940c^2d - 2054867641020c^2 \\
&\quad + 706181383584cd^2 + 151980984216cd + 1132596902196c \\
&\quad + 1884468968d^3 + 36794420832d^2 - 30413540316d + 6898957820.
\end{aligned}$$

B.2 Degree-order and lexicographic summaries

The 27 standard monomials in the degree-order quotient are

$$\begin{aligned}
&1; \quad d, d^2, d^3; \quad c, cd, cd^2, c^2, c^2d; \\
&b, bd, bd^2, bc, bcd, bc^2, b^2, b^2d, b^2c; \\
&a, ad, ad^2, ac, acd, ab, abd, a^2, a^2d.
\end{aligned} \tag{22}$$

They give the exact degree counts (1, 4, 10, 12, 0). The degree-order Gröbner basis has 21 rows, while the lexicographic shape has the four rows in equation (6); both quotient dimensions are 27. The independent checker reduces the original four equations and the four back-substituted numerators to zero.

B.3 Primitive eliminant

Write

$$g(d) = \sum_{i=0}^{27} g_i d^i.$$

The polynomial has content one and positive leading coefficient. The exact coefficient vector is:

i	g_i
0	-449439861345026126629260545097578647925157650368826090566763053593125389 687879139376966065914503487965224614417834357220517357813760000
1	100508745493009963936627007900913141901977978642287633678805892031802369 3134329099887650746185243453901519542697433222020561360650240000
2	-194470975657134669642176520626057624765129158726333245208006811729512737 3092548718032347250347259224021797092604341757295981232128000000
3	409598847830221610040865027146992733531524448239690730342977784711311504 4721049200473897440014624060709633385776097845568475590819840000
4	-438496120422004679441906541326009740716301458603342673708792688625584744 7234470211881456411655007340530876271251323151671360285573120000
5	656253833747163586353252199378775908258689874548471168271234546899608095 0983746003303177011231225201488210891048743198806834659459072000
6	-629470727433251494617173429194430962733153399960255703648600480642924711 5491176388244983999707946480020558142636990742037916071886848000
7	574945689844136142923590281629695780970464801998655246255913944304064492 1332577499053588818656195462705941543019918146002932745830400000
8	-609222183103198680754585163218836952023646005616556310017969528519556005 5183468215116095902293582943218418032018478826994564264558592000
9	441937878831718728704886727794012844857842089015504059036388068958496590 7344008427905768621893181016601521275723986434887264926760960000
10	-537034790909280469005370260009111941400337485535004926748393352987690784 6864607374920077439417717594136212163713650578304020668058828800
11	533007329206933304417306263318007104956058964838621904057538323710506566 2849809940522601705263330345135853744733457011706544175338291200
12	-612733607154138854768039557209053683751215924962617414591265403852600177 5510724161697318383174498266865978864469468443584043795345408000
13	680906319717763274346715254945850214529812898081848227987146957896389397 3547802403207903040336363618651053454880425435604774858553548800
14	-671740586361222439981721688001166773348309867826882004062542128750483297 1710737498474592388678424961411587655595975725978325624349798400
15	642522981842368833627449732347505129496322098341294868603290645877977581 7894231628828944794618937844084186401539249411491403946603550720
16	-526462561383939646416368808264033258080227295013536020453366574343028206 5601187769111491848740989960560194411766488608566820412255278080
17	426764625835676711234654957226535357427406366692376155556275017368131278 6104795465497625120129893558839216762193269052253387379841049600
18	-28538321163233553985648482758829365394328966941825684605675995055385372 2047396287717936315106929177141413980211305204291388268616675840
19	193100838689956488058063115432441155143061754467790489753594252701030356 5732141162598321141628122420367138574645844007801758973751468160
20	-113294674877994243406041844835341026671589499440703595811059993040854711 2390042644445558543920386284685438145840010008278530278600489792
21	566264320342732819404312868763534150585415112436068112670140711406290160 899876792798418105875511842395537344101999727313243053223672608
22	-325509876485852528549462697104196670393147532384334003583342408236849064 318706185517859610777325419277009260979341906384550981039991360
23	110958601116148422752293358737855278642943710056492833078145447498563053 721811177199596053680417151748718105278513864909993467518548640
24	-554833221510085519244780565955669808201432001163751952160454363462409629 32386410473707774409190009533790761714727560223103875184399540
25	163259626802836296016087798611540015020560322738225132037621227831729497 99107571227211453024532912761192870274438295117740022622649634
26	-367296984146469984546277935470299606173952970347040223651596495680214711 9332151397926754772909464298373324521537413520408849192062561
27	136481349183280799375507087695608690382278535977841146768926254878891353 7399544245964424834360406200467338436153217360729270457864920

The back-substitution arrays in equation (6) are substantially larger than the eliminant. They are retained without truncation in the canonical machine-readable certificate. Their mathematical use in the paper is fully replayable without trusting an FGLM label: the checker performs the four cleared substitutions into the displayed f_i and recomputes zero remainders modulo the coefficient vector above.

C Complete modular factors and Weyl-lattice data

C.1 Complete monic factors

All coefficients below are taken in $\mathbf{F}_p$. In each display the listed scalar is the factorization unit and all factors occur with multiplicity one.

$p = 7$.

$$\bar{g} = 2 \prod_{j=1}^7 \phi_{7,j},$$

where

$$\begin{aligned} \phi_{7,1} &= d^3 + 5d^2 + d + 3, & \phi_{7,2} &= d^3 + 3d^2 + 4d + 3, \\ \phi_{7,3} &= d^3 + 3d^2 + 5d + 4, & \phi_{7,4} &= d^3 + d^2 + 3d + 5, \\ \phi_{7,5} &= d^3 + 5d^2 + 6d + 5, \\ \phi_{7,6} &= d^6 + 3d^5 + 5d^4 + 5d + 3, \\ \phi_{7,7} &= d^6 + 2d^5 + 2d^4 + 3d^3 + 5d^2 + 5d + 6. \end{aligned}$$

$p = 19$.

$$\bar{g} = 8 \prod_{j=1}^5 \phi_{19,j},$$

where

$$\begin{aligned} \phi_{19,1} &= d + 7, \\ \phi_{19,2} &= d^4 + 7d^3 + 12d^2 + 12d + 11, \\ \phi_{19,3} &= d^4 + 14d^3 + 17d^2 + 4d + 15, \\ \phi_{19,4} &= d^6 + 11d^5 + 17d^3 + 6d^2 + 7d + 17, \\ \phi_{19,5} &= d^{12} + 4d^{11} + 4d^{10} + 14d^9 + 6d^8 + 17d^7 \\ &\quad + 15d^6 + 10d^5 + 14d^4 + 12d^3 + 13d^2 + 11d + 4. \end{aligned}$$

$p = 29$.

$$\bar{g} = 26 \prod_{j=1}^5 \phi_{29,j},$$

where

$$\begin{aligned} \phi_{29,1} &= d + 12, \\ \phi_{29,2} &= d^2 + 27d + 3, \\ \phi_{29,3} &= d^8 + 5d^7 + 20d^6 + 21d^5 + 25d^4 + 22d^3 + 4d^2 + 3d + 9, \\ \phi_{29,4} &= d^8 + 4d^7 + 6d^6 + 27d^5 + 3d^4 + 2d^3 + 23d^2 + 8d + 11, \\ \phi_{29,5} &= d^8 + 12d^7 + 18d^6 + 16d^5 + 21d^4 + 7d^3 + 11d^2 + 6d + 13. \end{aligned}$$

$p = 37$.

$$\bar{g} = 29 \prod_{j=1}^5 \phi_{37,j},$$

where

$$\begin{aligned}
\phi_{37,1} &= d^2 + 28d + 28, \\
\phi_{37,2} &= d^5 + 3d^4 + 17d^3 + 3d^2 + 2d + 19, \\
\phi_{37,3} &= d^5 + 34d^4 + 16d^3 + 22d^2 + 34d + 23, \\
\phi_{37,4} &= d^5 + 12d^4 + 30d^3 + 29d^2 + 27d + 26, \\
\phi_{37,5} &= d^{10} + 24d^9 + 23d^8 + 3d^7 + 2d^5 + 23d^4 + 5d^3 + 34d^2 + 30d + 5.
\end{aligned}$$

Direct multiplication reproduces the 28-entry coefficient vector of g modulo each prime, and every derivative gcd has degree zero. The prime-37 conclusion is deliberately split into four checks: smooth surface reduction; nonzero leading coefficient and squarefree eliminant; complete factor multiplication with cycle type $(2, 5, 5, 5, 10)$; and exclusion of that type from the Coxeter-even kernel.

C.2 Picard reflection model

In the ordered basis $(H, E_1, \dots, E_6)$, the intersection matrix is

$$J = \text{diag}(1, -1, -1, -1, -1, -1, -1),$$

and the anticanonical vector is

$$(3, -1, -1, -1, -1, -1, -1).$$

The six simple-root rows are

$$\begin{pmatrix}
0 & 1 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 \\
1 & -1 & -1 & -1 & 0 & 0 & 0
\end{pmatrix}.$$

Each row has square -2 , and its reflection preserves J , the anticanonical vector, the 27 line classes in equation (16), and their intersection matrix.

The exhaustive enumeration ledger is:

test	exact result
reflection-group order	51840
kernel of reflection determinant	25920
determinants of the six simple reflections	$(-1, -1, -1, -1, -1, -1)$
orbit of one line class	27
target cycle type	$(2, 5, 5, 5, 10)$
target-type count	5184
target-type count inside determinant kernel	0
target-type count outside determinant kernel	5184
ordinary S_{27} -odd elements in the full group	0
rank of common Picard fixed space	1

The last row is the machine input to, not the output of, the Hochschild–Serre argument. The arithmetic Picard number is derived in section 6; it is not a separately machine-computed lattice rank.

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